SMN2-RescueDiff: An Auditable Autoregressive Generation and RNA-Decoy Docking Pipeline for *SMN2* Splicing-Modifier Prioritization

Saksham Srivastava

***Abstract*—Spinal muscular atrophy can be treated by increasing inclusion of exon 7 in *SMN2* transcripts, but computational prioritization of RNA-binding small molecules requires both target screening and difficult off-target controls. This study introduces SMN2-RescueDiff, an auditable pipeline that couples unconditional autoregressive SELFIES generation, chemical filtering, five-conformer target docking, HuRDA-derived human RNA decoy docking, and multi-factor candidate ranking. From 150,000 training rows, 149,534 valid unique ChEMBL/ZINC molecules were used; 49,660 unique generated strings were emitted, 40,477 passed chemical filters, and 1,000 were exported for docking. The target campaign completed all 2,500 planned jobs for 500 molecules and five fixed *SMN2* RNA conformers. The staged decoy campaign completed 14,259 non-dry-run AutoDock Vina jobs and yielded a uniform 147×97 molecule–decoy matrix without score imputation. All 40 leading candidates had stronger mean target scores than their own property-matched control medians; the mean enrichment was 0.729 kcal/mol and exceeded every one of 2,000 random cohorts. This demonstrates reproducible target-score enrichment within the generated population and successful operation of the staged screening architecture. The same audit also identified an important boundary: all conservative target–decoy gaps were negative, and redocking of four experimental RNA–ligand complexes recovered a sub-2 Å rank-1 pose only for branaplam. Lower-RMSD modes were nevertheless sampled for SMN-CX and risdiplam, motivating RNA-aware reranking. The contribution is therefore a reproducible candidate-prioritization and validation framework, not evidence that any generated molecule binds RNA, rescues exon 7 inclusion, or has therapeutic activity.**

***Index Terms*—Autoregressive molecular generation, molecular docking, RNA, spinal muscular atrophy, reproducibility, virtual screening.**

I. Introduction

SPINAL Muscular Atrophy (SMA) patients experience degeneration of motor neurons, resulting in debilitating muscle atrophy and eventually full paralysis. Muscle atrophy from SMA also causes difficulty breathing and swallowing due to degeneration of throat muscles [1]. SMA is a neuromuscular disorder caused by insufficient functional survival motor neuron protein, most often due to chromosomal deletion of the Survival Motor Neuron 1 (*SMN1*) gene. Three drugs have been approved for SMA: risdiplam, an orally administered small molecule; nusinersen (Spinraza), an antisense oligonucleotide requiring repeated intrathecal administration; and onasemnogene abeparvovec (Zolgensma), a single-dose gene therapy. Current treatments do not restore mobility already lost to motor neuron degeneration, and small-molecule splicing modifiers carry the risk of off-target splice modulation [1]. The closely related *SMN2* gene can partially compensate for the disruption of *SMN1*, but inefficient inclusion of exon 7 during pre-mRNA splicing limits production of stable full-length SMN protein. Thus, *SMN2* offers a therapeutic route to increasing functional SMN protein [1]. The potential to restore patient mobility makes *SMN2* splicing an important therapeutic target, and the success of small-molecule splicing modifiers has shown that RNA processing can be pharmacologically altered to increase functional SMN expression [2]. In the SPR1NT trial, presymptomatic infants treated with onasemnogene abeparvovec within six weeks after birth achieved substantially better developmental outcomes than expected from the natural history of SMA [3]. However, RNA drug targets present a challenging drug-design problem because RNA structures are dynamic, highly charged, and able to form binding environments that can be difficult to distinguish computationally from unrelated RNA sites. RNA-targeted drug discovery must account for RNA structure, ligand recognition, and the difficulty of establishing selective interactions with drug-like small molecules [4], [5], [6]. These challenges have motivated growing interest in RNA-targeted small-molecule discovery and machine-learning methods for identifying RNA-binding sites and predicting RNA–ligand interactions [6].

Generative artificial intelligence offers a way to explore chemical space beyond pre-existing compound libraries by proposing new, chemically plausible molecules. Recent RNA-focused computational studies have addressed small-molecule binding-site prediction as well as broader machine-learning strategies for RNA-targeted drug design [7], [8]. Previous work has demonstrated the use of deep reinforcement learning and other generative approaches for de novo molecular design and molecular-property optimization [9], [10]. Deep reinforcement learning has been used to generate molecular structures and steer them toward specified design objectives [9], [11]. Reinforcement-learning-guided graph diffusion has also been investigated for generating molecules with improved predicted metabolic stability [10]. However, the ability to generate chemically plausible molecules does not, by itself, demonstrate

S. Srivastava is with Saint Francis High School, Mountain View, CA, 94040, United States (e-mail: saksham.shubh2010@gmail.com). ORCID ID: 0009-0007-9463-6294.

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meaningful interaction with a biological target. This is an especially important concern in RNA-focused models, where dataset biases or easily learned physicochemical differences between positive and negative examples can lead to shortcut learning rather than genuine recognition of target-specific interaction patterns [12]. Models developed for protein-targeted drug discovery cannot automatically be assumed to perform equally well for RNA targets, which require evaluation suited to RNA–ligand interactions [13]. Accordingly, rigorous RNA-targeted computational discovery requires not only molecular generation but also biologically relevant negative controls, structural screening, selectivity evaluation, and interpretable ranking of candidate molecules.

To address these challenges, this study developed SMN2-RescueDiff, a computational proof-of-concept framework in which molecular generation and biological target evaluation are treated as distinct but connected stages. The pipeline first generates and chemically filters candidate small molecules, then evaluates them against an independently defined *SMN2* exon 7/U1 splicing-related RNA target using multiple target conformations, HuRDA-derived human RNA decoys, and additional off-target counter-screens. Prioritization is performed using the Splicing Rescue Selectivity Index (SRSI), which integrates target-docking evidence, RNA counter-screening, splicing-related geometric proxies, chemical quality, and off-target penalties. The main objectives were to determine whether this staged framework could reproducibly move from molecule generation to an auditable shortlist, whether target-score enrichment remained after property matching and random-cohort controls, and whether candidates with favorable average separation retained separation from the single strongest decoy. The generator itself was deliberately unconditional; *SMN2* specificity entered only during docking and ranking.

Docking was performed with AutoDock Vina. Its scoring function was parameterized on protein–ligand complexes, and comparative assessments report that pose-prediction accuracy for RNA–ligand complexes is substantially lower than for protein targets and that current docking methods predict RNA binders unreliably [14], [15]. Vina was chosen here for throughput and reproducibility across a 15,000-job planned counter-screen rather than for demonstrated RNA accuracy. Consequently, docking values below are reported as comparative Vina scores, not binding free energies. An RNA-targeted generative-chemistry study has combined molecular design with compound synthesis and experimental testing, underscoring the further validation required after computational prioritization [16].

II. Materials and Methods

SMN2-RescueDiff is a modular computational biology pipeline designed to prioritize candidate small molecules for predicted *SMN2* exon 7 splicing rescue. The pipeline is organized into six modules that generate reproducible output files as inputs for later stages (Fig. 1): *SMN2* target definition, Human RNA Decoy Atlas construction, AI-based molecule generation, docking and off-target counter-screening, splicing-rescue scoring and final candidate ranking, and final virtual screening. This project is fully computational.

A. Module 1: SMN2 Target Definition

The first module defined the *SMN2* exon 7 splicing-related RNA target region, creating a reproducible reference target for all later stages. Because *SMN1* and *SMN2* are highly similar genes, the pipeline did not automatically select a target transcript when multiple *SMN2* transcript candidates were present. Instead, an explicit *SMN2* transcript ID was required in the configuration file before execution continued. Requiring the correct transcript reduced the risk of extracting the wrong paralog, transcript, or exon region.

The module took two input files: a human genome FASTA file and a matching GTF annotation file. The genome FASTA provided the genomic sequence, while the GTF annotation file provided gene, transcript, exon, strand, and coordinate information. The pipeline searched the annotation file for *SMN2*, enumerated transcript candidates, and required the selected transcript ID to be written into the configuration file.

After transcript selection, the module extracted the *SMN2* exon 7 region and nearby splicing-related target windows. These windows included the exon 7 sequence, the $5'$ splice-site region, the $3'$ splice-site region, and additional local target windows of 21, 51, and 101 nucleotides. The 21 nt core target window was used for precise target mapping. The 51- and 101-nt windows were used to characterize the sequence context from which the hardest decoys were subsequently selected. The $5'$ and $3'$ splice regions and the 21 nt window are the regions the generated molecules were intended to target; the $5'$ splice region is the exon-intron boundary and the $3'$ splice region is the spliceosome acceptor. Extracted genomic sequences were converted to RNA by substituting uracil for thymine, and the reverse complement was taken only when the GTF strand field indicated that *SMN2* lies on the minus strand, so that the extracted sequence always corresponds to the sense transcript.

Module 1 also calculated a target feature profile for each *SMN2* window, designed to describe the target region in a form usable for bias-matched RNA decoy construction. The calculated features included sequence length, nucleotide composition, A/U/G/C frequencies, GC content, purine fraction, pyrimidine fraction, dinucleotide counts, 3-mer counts, simple splice-site-related scores, U1 complementarity score, splice-boundary information, and positional distances relative to exon 7. This profile constitutes the *SMN2* target "fingerprint," which the Human RNA Decoy Atlas then used to identify human RNA motifs resembling the *SMN2* target in sequence and splicing-related features, so that later comparisons were made against hard rather than trivially distinguishable negative controls. A metadata record captured the project name, module name, selected gene, selected transcript, input file paths, created target windows, warnings, and limitations, so that downstream modules could trace exactly how the *SMN2* target was defined.

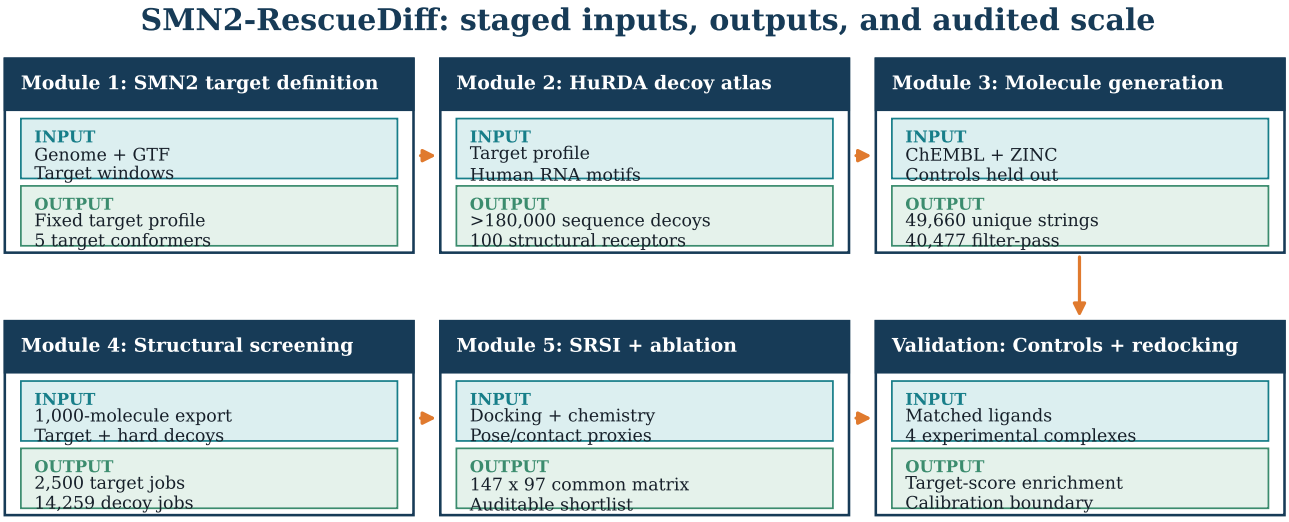


Fig. 1. Input-output diagram. *SMN2* feature calculation and HuRDA RNA-decoy construction establish the biological reference for screening, while molecular generation independently produces chemically filtered candidates. These streams converge in target docking and off-target counter-screening. Docking, selectivity, chemical-quality, and splicing-related proxy features are integrated through the Splicing Rescue Selectivity Index (SRSI), followed by benchmark comparison and candidate prioritization. Arrows indicate transfer of computational inputs and outputs.

B. Module 2: HuRDA Human RNA Decoy Atlas

The second module used the Module 1 outputs as reference inputs to construct an RNA decoy dataset. Its purpose was to test whether generated molecules bind selectively to the *SMN2* exon 7 target or also bind similar RNA motifs elsewhere in the human transcriptome.

The atlas was specified at a transcriptome scale of over 180,000 RNA motifs across eight categories: human 5′ splice-site decoys, human 3′ splice-site decoys, *SMN2*-like hard decoys, exonic regulatory windows, intronic regulatory windows, scrambled *SMN2* controls, mutated *SMN2* controls, and random bias-matched transcriptome windows. It is important to state the scope of what was actually used: this sequence-level atlas supplied the feature space from which decoys were selected, but only 100 decoy sequences were converted into three-dimensional receptors and docked. All structural selectivity results reported in this paper therefore rest on a 100-receptor panel, not on the full sequence atlas, and the sequence atlas contributes selection criteria rather than screening evidence.

For every decoy motif, the same general feature types calculated in Module 1 were recalculated, including length, nucleotide composition, GC content, k-mer similarity, splice-site score, U1 complementarity, and similarity to the *SMN2* target profile. These sequence and splice-related properties were used for bias matching; they were not treated as experimentally measured RNA accessibility or secondary structure. This allowed selection of hard, bias-matched decoys rather than random easy controls, reducing the risk that later screening would be evaluated only against trivially different sequences.

C. Module 3: AI Molecular Generation

Module 3 generated candidate small molecules for docking and screening. The quality-source manifest supported ChEMBL, ZINC, R-BIND 2.0 and SMMRNA inputs, but the audited run loaded molecules only from ChEMBL and ZINC: 74,558 and 74,976 valid unique records, respectively. The configured R-BIND and SMMRNA paths contributed zero rows and therefore did not provide RNA-ligand bias in this experiment. ChEMBL provides curated drug-discovery data, while ZINC20 provides a large collection of purchasable molecular structures for ligand discovery [17], [18]. Risdiplam and branaplam were maintained as held-out benchmark controls and were not used for training. The combined library was validated with RDKit, canonicalized, deduplicated, and expanded through source weights to 224,092 training exposures while retaining 149,534 unique molecules. A baseline SMILES n-gram generator and a character-level GRU generator were implemented, but no baseline performance comparison is claimed here; every molecule carried forward in this study came from the Transformer generator.

Generated SMILES strings were checked using RDKit, which determined whether each string could be parsed into a chemically valid molecular graph. Invalid SMILES were rejected, and valid molecules were canonicalized and filtered on computational chemical properties. The chemical filters included molecular weight, LogP, hydrogen bond donors, hydrogen bond acceptors, rotatable bonds, formal charge, ring count, aromatic ring count, QED drug-likeness, Lipinski rule compliance, PAINS alerts, novelty, and diversity. PAINS substructure alerts were used as screening flags for possible assay-interference liability rather than as experimental evidence that a candidate interferes with an assay [19]. Quantitative estimate of drug-likeness (QED) was included as a calculated descriptor of molecular properties associated with drug-like compounds [20].

1) Autoregressive molecular generation: The production generator was an autoregressive Transformer operating on

SELFIES tokens. After cleaning, canonicalization, and duplicate removal, the SMILES were converted to SELFIES, an alternative string representation of molecular structure. Each SELFIES string was split into molecular symbols, mapped to token identifiers, and supplied with start and end tokens. The Transformer learned to predict the next SELFIES token from the preceding tokens using causal self-attention. SELFIES was selected because its grammar is designed to produce valid molecular structures from valid SELFIES strings [21].

The architecture comprised six layers, eight attention heads, 384-dimensional embeddings, sinusoidal positional encodings, a 1,536-dimensional feed-forward network, and a dropout rate of 0.1. Training used AdamW with a learning rate of 0.0005, a weight decay of 10^{-5}, gradient clipping at a norm of 1.0, a batch size of 128, and next-token cross-entropy loss excluding padding positions, for up to 10 epochs with early stopping at a patience of six epochs. A random seed of 42 was used, and 10% of the tokenized sequences were reserved for validation.

Generation began with a start token and proceeded one SELFIES token at a time until an end token was sampled or 160 sampling steps had been completed. Sampling used a temperature of 0.78, top-k filtering with $k = 50$, and top-p filtering with $p = 0.95$. The sampled token identifiers were reconstructed into SELFIES strings and converted to SMILES using the SELFIES decoder. These decoded SMILES were parsed, sanitized, and canonicalized with RDKit before duplicate removal, chemical-property filtering, and candidate prioritization. SELFIES therefore served as the model's training and generation representation, while SMILES served as the format for subsequent molecular processing and export. *SMN2* specificity was assessed downstream through target-conformer docking, HuRDA RNA counter-screening, and SRSI ranking; the generator received molecular tokens without *SMN2* sequence conditioning.

The autoregressive objective is given in (1):

$$p_\theta(x_{1:T}) = \prod_{t=1}^{T} p_\theta(x_t \mid x_{<t}),$$
$$\mathscr{L}_{\mathrm{CE}} = -\frac{\sum_{n,t} m_{nt} \log p_\theta(x_{nt} \mid x_{n,<t})}{\sum_{n,t} m_{nt}}, \tag{1}$$

where x_{nt} is token t of training molecule n and m_{nt} excludes padding tokens.

D. Module 4: Docking and Off-Target Counter-Screening

This module computed comparative docking scores and selectivity proxies for candidate molecules at the *SMN2* exon 7 splicing-related RNA target and selected human RNA decoys from HuRDA.

Molecular docking requires a three-dimensional receptor structure. A hard-decoy panel selected from HuRDA was converted into predicted 3D RNA receptor structures using RNA Composer, then prepared as PDBQT receptors for structural counter-screening. Candidate–receptor poses and docking scores were generated using AutoDock Vina, whose original search and scoring approach and subsequent software extensions have been described in the literature [22], [23]. All decoy docking scores were non-dry-run Vina outputs, and pose files were generated for molecule-decoy pairs.

The docking module calculated *SMN2* target docking scores, decoy docking scores, and target-to-decoy selectivity metrics. Because more negative Vina values are treated as more favorable by the scoring function, selectivity was computed so that a larger positive gap corresponds to a more favorable target score relative to decoys. Candidates showing favorable *SMN2* scores and less favorable decoy scores advanced to the splicing-rescue scoring module.

E. Module 5: Splicing Rescue Selectivity Index and Ranking

Module 5 combined docking and chemical-property results into the SRSI. For each molecule, the HuRDA decoy score was computed as a biologically weighted mean across the receptors that were successfully scored. Subtracting the mean *SMN2* conformer Vina score from this quantity yielded the weighted selectivity gap; because more negative Vina scores are more favorable within the scoring function, a larger positive gap corresponded to a more favorable comparative score for *SMN2*. A conservative gap was also calculated by comparing the strongest-scoring decoy with the molecule's least favorable *SMN2* conformer, and decoy coverage recorded the fraction of planned comparisons that produced usable scores (2). Failed docking jobs were not assigned estimated scores.

For successfully scored decoys $\mathscr{D}_i$, with biological weights w_j:

$$\overline{E}_i^{D,w} = \frac{\sum_{j\in\mathscr{D}_i} w_j E_{ij}^D}{\sum_{j\in\mathscr{D}_i} w_j},$$
$$\Delta_i^w = \overline{E}_i^{D,w} - \overline{E}_i^T, \tag{2}$$
$$\Delta_i^{\mathrm{conservative}} = \min_{j\in\mathscr{D}_i} E_{ij}^D - E_{i,\mathrm{worst}}^T,$$
$$\mathrm{Coverage}_i = \frac{|\mathscr{D}_i|}{M},$$

where M is the planned number of decoys. Weights w_j were read from the decoy annotation; decoys without a recorded weight defaulted to $w_j = 1$, which is a scoring convention rather than a statement of biological importance.

1) Splicing-rescue proxy: The splicing-rescue proxy combined weighted target-docking, splice-region and U1-related, and contact-support components, together with pose-based support when usable pose evidence was available or feature-based support when it was not. Penalties for missing pose evidence, splice-site occlusion, and off-target risk were then

applied; the result was scaled by the configured denominator and constrained to the interval $[0, 1]$:

$$R_i = \mathrm{clip}_{[0,1]}\left[\begin{array}{l}\frac{1}{D}\Big(\sum_{k\in\mathscr{P}} a_k u_{ik} + 2m_i F_i \\ \qquad + 1.5\,(1-m_i)V_i \\ \qquad - b_C C_i - b_O O_i - b_m m_i\Big)\end{array}\right], \quad (3)$$

in which u_{ik} are positive support components (target-docking, landmark/U1 support and contact proxies); F_i is feature-based support; V_i is the pose-support score; m_i indicates missing or unusable pose evidence; C_i is the splice-site occlusion penalty; O_i is off-target risk; and a_k, b_C, b_O, b_m are configured weights. The implementation uses

$$D = \max\left(10^{-9}, \sum_{w\in\text{configured rescue weights}} |w| + 2\right).$$

The configured positive weights were 2.0 for beneficial-landmark proximity, 2.0 for U1 support, 1.5 for exon-7-boundary support, 1.5 for target score, 1.0 for contact density, 0.8 for the hydrogen-bond proxy, and 0.7 for the stacking proxy. Penalty weights were 2.0 for occlusion, 1.5 for off-target risk, and 0.5 for missing pose evidence. A usable pose contributed $1.5V_i$; a missing pose contributed $2F_i$.

Contact features were atom-pair proxies with fixed cutoffs: 4.5 Å for a close contact, 3.5 Å for a hydrogen-bond-like pair, and 5.0 Å for a carbon–carbon stacking-like pair. For N_i ligand atoms, contact density was $\min(1, n_{\mathrm{close}}/(4N_i))$; hydrogen-bond and stacking scores were $\min(1, n/(0.5N_i))$; and $V_i = \mathrm{clip}_{[0,1]}(0.45d_i + 0.30h_i + 0.25s_i)$. The implementation did not map every receptor atom to a nucleotide landmark. If a U1 or 5′-site label was present, beneficial contacts were assigned as $\lfloor 0.75n_{\mathrm{close}} \rfloor$; if donor or acceptor labels were present, caution contacts were assigned as $\lfloor 0.35n_{\mathrm{close}} \rfloor$, and $C_i = n_{\mathrm{caution}}/\max(1, n_{\mathrm{close}})$. These are explicit engineering proxies, not experimentally observed contacts or mechanistic splicing measurements.

2) SRSI standardization and integration: For the final SRSI, continuous features were standardized across the scored candidate cohort and combined using the configured weights. The Lipinski-pass and PAINS-alert indicators were included without standardization. For a scoring feature x:

$$z_i(x) = \frac{x_i - \mu_x}{\sigma_x}, \qquad \mu_x = \frac{1}{n}\sum_i x_i,$$

$$\sigma_x = \sqrt{\frac{1}{n}\sum_i (x_i - \mu_x)^2}.$$

The complete scoring equation is

$$\begin{aligned} S_i = {} & w_B z_i\big(-E^T\big) + w_R z_i(R) + w_\Delta z_i(\Delta^w) + w_Q z_i(Q) \\ & + w_L I_i^L + w_P z_i(P) - w_O z_i(O) - w_A z_i(A) \\ & - w_H I_i^H - w_C z_i(C), \end{aligned} \quad (4)$$

with $-E^T$ target-docking strength ($w_B = 1.0$); R rescue proxy ($w_R = 2.0$); Δ^w weighted selectivity gap ($w_\Delta = 1.5$); Q QED ($w_Q = 0.7$); I^L Lipinski-pass indicator ($w_L = 0.4$); P pre-docking score ($w_P = 0.5$); O off-target risk ($w_O = 1.5$, subtracted); A lower-is-better SA score ($w_A = 0.5$, subtracted); I^H PAINS-alert indicator ($w_H = 1.0$, subtracted); and C occlusion penalty ($w_C = 1.2$, subtracted).

F. Module 6: Final Virtual Screening Protocol

Module 6 specified the staged evaluation applied to the generated candidates. Rather than relying on a single docking score, candidates passed through target-conformer screening, RNA counter-screening, ensemble-aware selectivity analysis, additional off-target screening, and benchmarking against established *SMN2* splicing modifiers, and were required to satisfy several independent computational criteria before advancement.

1) Target-conformer robustness screening: Five fixed *SMN2* target conformers were prepared and used as a receptor ensemble. A pilot 10×5 experiment preceded full screening of 500 candidate molecules against all five conformers. For each molecule the pipeline calculated mean, median, best, and worst target Vina scores, standard deviation across conformers, and total target-score range, and prioritized candidates by more favorable worst-conformer score, more favorable mean score, and lower variation across the conformational ensemble. Molecules requiring lower-confidence rigid-tree repair during the target screen were excluded from standard advancement, preventing repaired docking artifacts from influencing selection. For $K = 5$ conformers:

$$\begin{aligned} \overline{E}_i^T &= \frac{1}{K}\sum_{c=1}^{K} E_{ic}^T, \qquad E_{i,\mathrm{worst}}^T = \max_c E_{ic}^T, \\ s_i^T &= \sqrt{\frac{1}{K-1}\sum_{c=1}^{K}\Big(E_{ic}^T - \overline{E}_i^T\Big)^2}. \end{aligned} \quad (5)$$

2) HuRDA RNA counter-screening: The 150 conformer-robust molecules were evaluated against 100 human RNA decoy receptors derived from the HuRDA atlas. To reduce computational load while preserving the complete receptor panel, the experiment was divided into two docking runs: 150 molecules against 53 receptors and 150 molecules against the remaining 47 receptors. This stage was designed to move beyond an absolute target score and evaluate comparative RNA docking behavior: a molecule receiving a favorable *SMN2* score but equally favorable scores on many unrelated human RNA structures is less compelling than one with a wider target–decoy score gap. Only successfully completed docking calculations were retained in the merged dataset; failed jobs were not assigned artificial or imputed scores.

3) Ensemble-aware selectivity analysis: Candidates were ranked lexicographically on five criteria in order: larger weighted mean selectivity gap, larger conservative selectivity gap, reduced predicted off-target risk, a more favorable mean *SMN2* score, and lower variation across the five *SMN2* conformers. A later criterion is consulted only when every

earlier criterion ties. The conservative selectivity calculation imposed a stricter test than comparison of average scores alone: it evaluated whether *SMN2* remained the preferred predicted target when the candidate's least favorable *SMN2* conformer was compared with its strongest apparent interaction among off-target RNAs. This reduced the 150-molecule cohort to the top 40 candidates.

4) Cross-target off-target screening: Each of the top 40 candidates was evaluated against 20 additional target-decoy receptors, producing 800 planned docking calculations, to identify candidates that maintained favorable *SMN2*-directed performance while minimizing strong predicted interactions with additional non-primary targets. This stage distinguished molecules that appeared promising primarily because of a favorable target score from molecules that retained a more favorable computational specificity profile across multiple target classes.

5) Benchmark comparison with known SMN2 splicing modifiers: Risdiplam and branaplam were processed under the same protocol as the generated candidates, because a credible computational prioritization pipeline should be able to recognize established *SMN2* splicing modifiers before predictions for newly generated molecules are interpreted. Benchmark compounds were compared on target engagement, selectivity, and developability. Developability was scored with desirability functions rather than raw min–max normalization, so that each value is interpretable against a fixed, prespecified acceptance window instead of against the other two compounds. For a preferred interval $[a, b]$ with author-set tolerances α and β,

$$d(x) = \begin{cases} 0, & x \le a - \alpha, \\ \dfrac{x - (a - \alpha)}{\alpha}, & a - \alpha < x < a, \\ 1, & a \le x \le b, \\ \dfrac{(b + \beta) - x}{\beta}, & b < x < b + \beta, \\ 0, & x \ge b + \beta, \end{cases} \tag{6}$$

and for one-sided criteria with target T and zero-credit bound L or U,

$$d_{\uparrow}(x) = \mathrm{clip}\left(\frac{x - L}{T - L}, 0, 1\right), \quad d_{\downarrow}(x) = \mathrm{clip}\left(\frac{U - x}{U - T}, 0, 1\right). \tag{7}$$

Per-metric desirabilities were aggregated as a weighted geometric mean,

$$D = \left(\prod_{m=1}^{M} d_m^{\omega_m}\right)^{1/\sum_m \omega_m}, \tag{8}$$

so that a metric scoring $d_m = 0$ drives D to zero. All anchors, tolerances, and weights are given in Table II; no parameter was fitted to the compared compounds.

The benchmark strategy evaluated more than docking score alone. Native-pose redocking was carried out against four experimentally determined RNA–ligand complexes, including risdiplam and branaplam, using five seeds, 20 retained modes per run, and symmetry-corrected heavy-atom RMSD. Pose recovery and native-contact overlap were analyzed separately from candidate ranking, as reported in Section III-H. Benchmark controls were not used for model training.

G. Stage C.5: Matched Ligand Decoy Controls

The HuRDA panel of Section II-B varies the receptor while holding the ligand fixed, and therefore cannot distinguish a candidate that engages *SMN2* specifically from one that docks well because of its size and physicochemical profile. Stage C.5 applies the complementary control: the receptor ensemble is held fixed and the ligand is varied, by comparing each ranked candidate with property-matched decoy ligands drawn from the same generated population.

Controls were selected from the 500-molecule target-ensemble population after removing the 40 ranked candidates and any exact structural duplicates, leaving a pool of 460. Nine descriptors were standardized across the combined candidate and control population—molecular weight, cLogP, hydrogen-bond donors and acceptors, tPSA, rotatable bonds, formal charge, QED and a synthetic-accessibility proxy—and each candidate was matched to its ten nearest controls by Euclidean distance in that standardized space. A Morgan fingerprint guard (radius 2, 2048 bits) excluded any control with Tanimoto similarity above 0.6 to its candidate, so that matched pairs share physicochemical profile without sharing scaffold. Controls could be reused across candidates, and reuse counts are reported. For candidate i with matched-control set $\mathscr{C}_i$, the target enrichment gap is

$$g_i = \underset{c \in \mathscr{C}_i}{\mathrm{median}} \, \overline{E}_c^{T} \; - \; \overline{E}_i^{T}, \tag{9}$$

so that $g_i > 0$ means the candidate receives a more favorable mean *SMN2* score than the median of its own matched controls. The 40 candidate matched sets were treated as the independent units. Confidence intervals were obtained by bootstrap over those sets (10,000 resamples), and significance by a random sign-flip permutation test on the candidate-level gaps (20,000 permutations), both under seed 61453. Stage C.5 performed no new docking: it reuses the five-conformer target results of (5), and no missing score was repaired or imputed.

III. Results

A. Generation and chemical triage

The audited generation run converted a broad chemical-language model into a manageable docking cohort (Table I). Of 150,000 source rows, 149,534 parsed as valid, unique molecules and 466 were rejected. Source weighting produced 224,092 training exposures without altering the unique-molecule record. Training ran for 10 epochs; final training and validation cross-entropy losses were 0.6287 and 0.5722, respectively.

The run requested 50,000 samples and emitted 49,660 unique generated strings. Every emitted string parsed with RDKit, while canonicalization left 49,489 unique molecular graphs. Chemical-quality filters retained 40,477 molecules and rejected 9,183. Of the retained set, 40,406 were novel under the run's approximate comparison with the first 20,000 canonical training molecules; this is not a full-corpus novelty

TABLE I
AUDITED MOLECULE-GENERATION AND SELECTION COUNTS

Stage	Observed count
Source rows inspected	150,000
Valid unique training molecules	149,534
Invalid source rows	466
Weighted training exposures	224,092
Unique generated strings emitted	49,660
Unique valid canonical molecules	49,489
Molecules passing chemical filters	40,477
Molecules rejected by filters	9,183
Candidates exported for docking	1,000
Recorded scaffolds in docking export	977

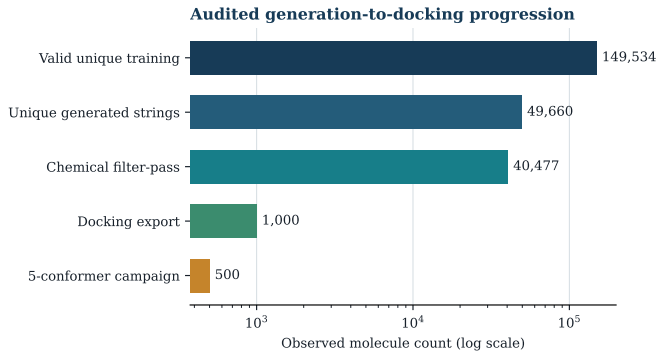


Fig. 2. Audited progression from source-library validation through generation, chemical filtering, and diversity-aware docking export. Counts are observed pipeline outputs, not expected or back-filled targets. The 500-molecule structural campaign was selected from the 1,000-molecule docking export.

estimate. Diversity-aware selection exported 1,000 candidates for docking, representing 977 recorded scaffold labels. Across 50,000 sampled fingerprint pairs, the mean and median Morgan Tanimoto similarities were 0.121 and 0.115, respectively, indicating that the exported cohort was not dominated by near-duplicate structures. A post hoc audit against the two held-out splicing controls found no filtered or exported candidate above the prespecified Tanimoto exclusion threshold of 0.70. The original run tracked branaplam alone; adding risdiplam to the audit did not rewrite the historical generation output.

B. Docking campaign and data completeness

The five-conformer *SMN2* target screen completed all 2,500 planned ligand–conformer jobs across the 500 generated candidates, producing a complete target matrix with no missing entries. The completed dataset was verified to contain 500 unique molecules, five unique conformers, no duplicate molecule-conformer pairs, and five completed conformer scores for every molecule. Two molecules required rigid-tree repair during ligand preparation; these are retained with a reduced-confidence flag, and their scores are reported but are not treated as having the same confidence as those of the remaining 498. This screen reduced the 500-molecule set to 150 conformer-robust candidates, which were frozen with cryptographic checksums to preserve the exact target-screening outputs used downstream.

The HuRDA counter-screen was designed as 150 molecules against 100 decoys and executed as two panels of 53 and 47 receptors, for 15,000 intended molecule–decoy pairs. Of these, 14,259 completed successfully. The 741 jobs that did not launch are fully attributable to unavailable docking inputs rather than to any docking outcome: three decoy receptors could not be prepared in PDBQT format, and three ligands could not be converted from their stored structures. Applying inclusion–exclusion yields $(3)(150)+(3)(100)-(3)(3) = 741$, where the final term removes the nine pairs missing both a ligand and a receptor. The preflight check reports 750 input problems for these 741 jobs, because those nine pairs are counted twice; the two figures are therefore consistent, not contradictory.

Because these failures arise from structural-input issues, no score was imputed for the affected pairs and no placeholder score was substituted. Global coverage is therefore $14{,}259/15{,}000 = 95.06\%$, with no duplicate molecule–decoy pairs in the merged table. The surviving data are not ragged: 147 of the 150 molecules retain an identical, complete 97-decoy panel, so every ranking comparison is made on a common decoy set. The remaining three molecules have zero decoy coverage, are labeled insufficient decoy evidence, and are excluded from ranking while their identities are preserved in the output. Each evaluable molecule is thus assessed on 97% of its intended panel. The cross-target screen completed all 800 planned 40×20 docking calculations and passed computational validation checks.

C. Selectivity profile of the leading candidates

The 147 evaluable candidates were ranked by the lexicographic rule of Section II-F3; this ensemble-aware selectivity stage is referred to as Stage C, and the matched-control audit built on its output as Stage C.5. Fig. 3 presents the ten highest-ranked candidates. Their weighted selectivity gaps ranged from 6.177 to 6.622 kcal/mol, indicating favorable average separation within the evaluated cohort. However, all conservative gaps were negative (-2.607 to -0.367 kcal/mol): for every candidate, at least one decoy scored more favorably than that candidate's least favorable *SMN2* conformer. None of the ten candidates met the prespecified thresholds for mean target Vina score, weighted selectivity gap, conservative gap, or off-target risk. The pattern is not confined to the tabulated ten. Across all 147 evaluable molecules the conservative gap is negative without exception, ranging from -3.235 to -0.367 kcal/mol with a median of -1.950 and an interquartile range of $[-2.304, -1.718]$, while the weighted gap over the same 147 spans only 5.274 to 6.622 kcal/mol. Favorable mean separation therefore coexists with a lack of worst-case separation in every case, and a pipeline reporting only the mean gap would have assigned all ten an unsupported selectivity interpretation.

The off-target risk index takes the identical value 0.657216 for five of the ten candidates and 0.657216 or 0.653608 for six. Such exact ties indicate that the index saturates at a ceiling imposed by its clipped component terms rather than resolving differences among these candidates. Because the ranking is lexicographic, this near-degeneracy in the third criterion has almost no influence on the final order, which is

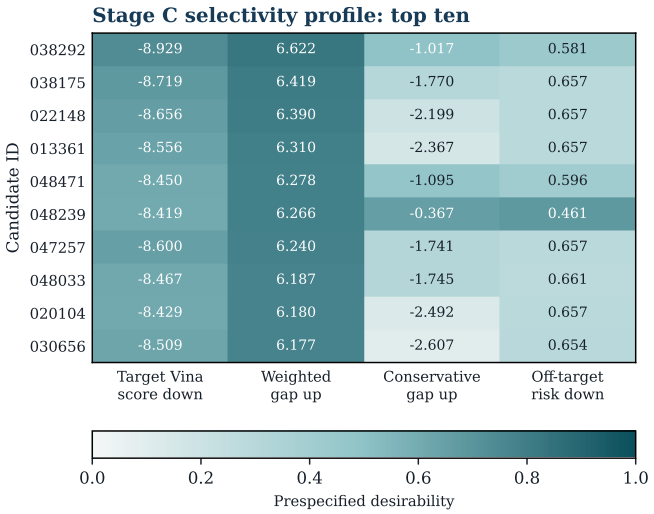


Fig. 3. Selectivity profile of the ten top-ranked candidates, ordered by the lexicographic rule of Section II-F3. Each cell reports the measured value and is shaded by its desirability $d \in [0, 1]$ with respect to the acceptance target for that metric, computed from (7) with the anchors and acceptance targets given in Table II. Arrows give the optimization direction: ↓ smaller-is-better, ↑ larger-is-better.

determined in practice by the continuous weighted gap alone. MolGen_048239 illustrates the consequence: it ranks sixth on the primary criterion, yet carries both the least negative conservative gap (-0.367 kcal/mol) and the lowest off-target risk index (0.461) of the ten. Under the stated ordering this profile confers no ranking advantage.

D. Benchmark comparison

The top-ranked generated candidate, MolGen_038292, was compared with risdiplam and branaplam on target scoring, RNA selectivity, and developability (Fig. 4). It showed the most favorable mean target Vina score (-8.929 versus -8.613 and -8.321 kcal/mol), the largest weighted selectivity gap (6.622 versus 6.309 and 6.071 kcal/mol), the lowest off-target risk index (0.581 versus 0.664 and 0.600), and the smallest fraction of decoys scoring more strongly than the target (0.093 versus 0.124 and 0.165).

MolGen_038292 was therefore competitive with the benchmark controls on several outputs of the same computational protocol, which supports its retention as a follow-up candidate. These are not clinical-superiority claims. First, all three compounds fall outside every author-defined acceptance window in the target and selectivity blocks. Because this includes two established splicing modifiers, the windows function as strict exploratory triage criteria rather than calibrated biological decision thresholds. Second, MolGen_038292 has a molecular weight of 504.0 Da against 401.5 and 393.5 Da for risdiplam and branaplam. AutoDock Vina scores scale with ligand heavy-atom count, so a 0.32–0.61 kcal/mol advantage over compounds roughly 100 Da smaller is of the magnitude expected from size alone, and ligand-efficiency normalization is required before any target-score advantage can be interpreted. Third, several of the candidate's physicochemical properties fall outside the preferred ranges, and its overall desirability

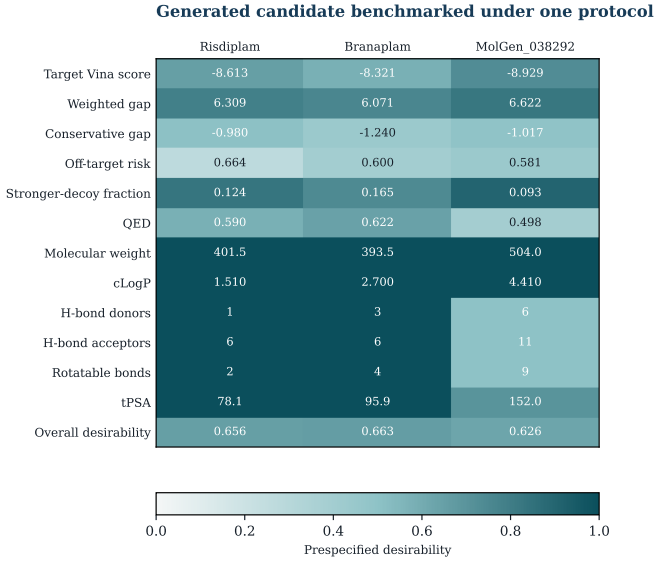


Fig. 4. Multi-objective benchmark of the top-ranked generated candidate against two clinical-stage *SMN2* splicing modulators, grouped into target scoring, selectivity and developability. Each cell reports the measured value and is shaded by its desirability $d \in [0, 1]$ from (6) and (7), with anchors and the acceptance window for each metric given in Table II. The bottom row is the overall desirability D of (8). Reference-compound descriptors are literature values (PubChem CID 118513932; branaplam free base, $C_{22}H_{27}N_5O_2$).

is 0.626, lower than risdiplam (0.656) and branaplam (0.663). The benchmark therefore identifies a computationally competitive generated molecule while also showing why the pipeline must retain developability, size, and calibration checks.

TABLE II
DESIRABILITY ANCHORS, TOLERANCES AND WEIGHTS USED IN FIG. 3 AND FIG. 4

Metric	Form	Parameters	ω
Mean target Vina score	$d_\downarrow$	$T = -10$, $U = -6$	3.0
Weighted selectivity gap	$d_\uparrow$	$L = 0$, $T = 8$	3.0
Conservative gap	$d_\uparrow$	$L = -3$, $T = 1$	2.5
Off-target risk index	$d_\downarrow$	$T = 0.30$, $U = 0.80$	2.5
Stronger-decoy fraction	$d_\downarrow$	$T = 0.05$, $U = 0.50$	1.5
QED	$d_\uparrow$	$L = 0.30$, $T = 0.80$	2.0
Molecular weight (Da)	d	$[220, 520]$, $\alpha = 70$, $\beta = 80$	1.0
cLogP	d	$[0, 4.5]$, $\alpha = \beta = 1.0$	1.0
H-bond donors	d	$[0, 5]$, $\alpha = 0$, $\beta = 2$	0.75
H-bond acceptors	d	$[2, 10]$, $\alpha = \beta = 2$	0.75
Rotatable bonds	d	$[0, 8]$, $\alpha = 0$, $\beta = 2$	0.75
tPSA (Å^2)	d	$[35, 140]$, $\alpha = 15$, $\beta = 40$	1.0

E. Matched ligand decoy control enrichment

Stage C.5 compared the 40 highest-ranked candidates with property-matched decoy ligands under the protocol of Section II-G. Matching assigned 400 candidate–control pairs drawn from 183 distinct control molecules, ten to each candidate. The match is close on physicochemical profile and distant on structure: the mean absolute standardized difference does not exceed 0.67 on any of the nine descriptors, while the largest candidate–control Tanimoto similarity observed is 0.338, with a mean of 0.124. No control exceeded the

0.6 similarity guard, so no relaxation of the threshold was required.

Every one of the 40 candidates received a more favorable mean *SMN2* conformer score than the median of its own matched controls (Fig. 5). The mean target enrichment gap of (9) is 0.729 kcal/mol, with a bootstrap 95% confidence interval of $[0.660, 0.802]$; the median is 0.686, with a bootstrap 95% confidence interval of $[0.620, 0.835]$. A random sign-flip permutation test on the candidate-level gaps returns $p \leq 5.0 \times 10^{-5}$, the resolution limit of 20,000 permutations, since no permutation reached the observed statistic; the paired effect size is $d_z = 3.15$.

This is the one comparison in the study in which the generated candidates separate from a control set, and it isolates a specific claim: the target scores are not explained by molecular size or drug-likeness, because the controls match on both. It does not establish *SMN2* selectivity, which is what the decoy receptor panel tests and what Section III-C reports as failing. Nor is the control set independent of the generator: the controls come from the same 500-molecule population and inherit its chemistry, so the comparison measures enrichment within that population rather than against an external library.

A further confound is that the controls are not independent of the selector either. The first lexicographic criterion is the weighted selectivity gap of (2), which a molecule raises by receiving a more favorable target score, so the 40 candidates were chosen partly for the quantity Stage C.5 then measures. Descriptor matching removes size and drug-likeness as explanations but cannot remove this one. To quantify it, the protocol of Section II-G was rerun on 2,000 cohorts of 40 molecules drawn at random from the 460 unselected molecules, each matched to ten controls from the remainder under identical rules, including the Tanimoto guard. Every cohort found a full complement of controls. The random cohorts are centered on zero enrichment (mean -0.049 kcal/mol, standard deviation 0.099, median -0.056), with a 99th percentile of 0.224 and a largest cohort mean of 0.367. The observed 0.729 lies 7.84 standard deviations above the null mean and above all 2,000 random cohorts ($p \leq 5.0 \times 10^{-4}$, the resolution limit of the draw), so the enrichment is not a sampling artifact.

That null varies the cohort at random, however, and therefore does not isolate selection by target score, which is what the first lexicographic criterion rewards. A second control addresses it directly: a cohort of the 40 molecules with the most favorable mean target score alone, with no decoy or selectivity term, matched under the same protocol. That cohort enriches at least as much as the ranked one (median 0.723 against 0.686 kcal/mol, mean 1.016 against 0.729, 40 of 40 positive in both), and shares 35 of its 40 members. The enrichment is therefore a consequence of selecting for favorable target scores. The matched controls establish that this is not an artifact of molecular size or drug-likeness, and the random null that it is not chance, but neither shows that the multi-objective rule contributes anything beyond what ranking on target score alone would deliver.

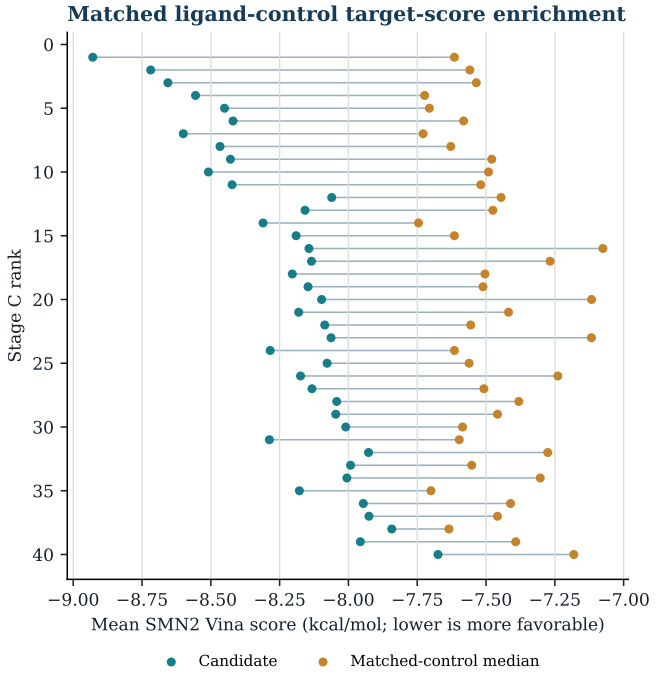


Fig. 5. Matched ligand decoy control enrichment for the 40 highest-ranked candidates, ordered by Stage C selectivity rank. Each row pairs one candidate with the median of its ten property-matched controls; the horizontal segment is the target enrichment gap of (9). More negative Vina scores are more favorable within this scoring function, so a candidate marker to the left of its control marker indicates enrichment. Controls were matched on nine standardized descriptors and constrained to Tanimoto similarity below 0.6, giving a largest observed similarity of 0.338. Scores are reused from the five-conformer target screen; Stage C.5 ran no new docking calculations.

F. Pose contact evidence

The pose-support and occlusion terms of (3) are computed from geometric contact features parsed from the docked pose and receptor coordinates. They were recomputed for the 147 evaluable molecules reported here, against the same five-conformer target poses used throughout, so the values below describe the ranked cohort rather than a separate one. All 147 scored in pose-based rather than feature-fallback mode.

The median molecule makes 1,631 close contacts, every one of them within the defined target region, of which 1,223 fall on beneficial splicing landmarks and 570 on caution landmarks. The median minimum distance to the U1 region is 1.653 Å (range 0.970 to 2.139). Hydrogen-bond and stacking proxy counts have medians of 197 and 522, spanning 107 to 323 and 323 to 716 respectively.

Two features of this distribution limit what the terms can contribute. The pose-support score V_i is at its ceiling of 1.000 for the median molecule and falls no lower than 0.928, and the splice-site occlusion penalty C_i spans only 0.349254 to 0.350000 across all 147 molecules, a range of 7.5×10^{-4}. A term with that range cannot order a cohort, so the occlusion penalty enters (4) with weight 1.2 while carrying almost no discriminating information. This is a property of the contact model as implemented, not of the molecules, and it holds on this cohort as it did on an earlier 200-molecule campaign, so it is a stable property of the scoring function rather than an accident of one candidate set.

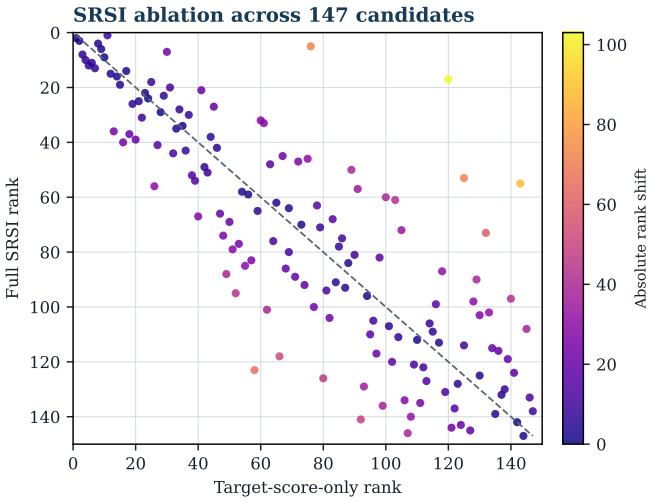


Fig. 6. Comparison of target-score-only and full-SRSI ranks for 147 candidates. Point color reports absolute rank shift and the diagonal marks no change. External activity data are required to determine whether the observed changes are beneficial.

G. SRSI ablation and rank stability

The final 147-molecule cohort was reranked under binding-only, binding-plus-selectivity, binding-plus-splicing-proxy, and full-SRSI configurations. Relative to the full ranking, Spearman rank correlations were 0.822 for binding only, 0.891 after adding selectivity, and 0.855 after adding the splicing proxy. Seven of the full-SRSI top ten also appeared in the binding-only top ten, eight appeared in the binding-plus-selectivity top ten, and seven appeared in the binding-plus-splicing-proxy top ten. The binding-only and full-SRSI top 40 shared 33 molecules. Across the full cohort, the median absolute binding-to-full rank shift was 14 positions, with a maximum of 103.

These results show that the composite score changes individual ranks and that selectivity contributes measurable ordering information (Fig. 6); they do not show that the full SRSI is more biologically predictive, because no independent activity labels were available for this cohort. The ablation therefore audits the behavior of the ranking rule rather than validating the configured weights.

H. Redocking validation of the docking protocol

Every quantity reported above derives from AutoDock Vina scores, so the protocol was tested against RNA–ligand complexes whose bound poses are experimentally determined. Four were used: 8R62 and 8R63, risdiplam and branaplam bound to the RNA duplex formed on 5′ splice-site recognition; 8R8P, the splicing modifier SMN-CX bound to the RNA helix formed on *SMN2* exon 7 5′ splice-site recognition; and 5KX9, a ribocil analogue bound to the FMN riboswitch, as an unrelated RNA–ligand positive control. Each ligand was rebuilt from SMILES, so no run began from the deposited conformer. The protocol—grid box, exhaustiveness 32, 20 modes, five seeds—was fixed and hashed before any RMSD was computed. Across 45 deposited receptor models and five seeds, all 225 planned jobs completed and 4,488 Vina modes were retained for symmetry-corrected analysis.

Top-ranked pose recovery was limited but informative (Table III). Branaplam met the predeclared rank-1 < 2 Å criterion: its top-scored mode was 1.307 Å from model 1 and 1.244 Å from the closest deposited ensemble member, reproducing 17 of 21 native contacts. Risdiplam, at the same site and under the same protocol, was not recovered at rank 1. Its top-scored mode was 12.587 Å from model 1 (11.923 Å ensemble minimum), shared none of 29 model-1 native contacts, and no seed placed a sub-2 Å pose first. The unrelated riboswitch control also failed at rank 1, at 6.173 Å with 27% contact recall, so the difficulty is not peculiar to the splice site.

The risdiplam rank-1 mismatch is diagnostically useful. A pose within 2.478 Å of the deposited ensemble *was* generated, but ranked fourteenth of twenty. Sampling approached the neighborhood of the correct answer; scoring did not prioritize it. The same pattern holds for SMN-CX, where the best mode of twenty reaches 1.856 Å but ranks last. Even branaplam's success is fragile: only one of five seeds ranked a sub-2 Å pose first. Deposited ensemble spread is 0.68 to 1.31 Å at the median, so the deviations exceed the experimental uncertainty by a wide margin.

IV. Discussion

The principal result is that SMN2-RescueDiff functioned as an end-to-end, auditable prioritization pipeline at a scale large enough to expose both candidate signals and method limitations. It transformed a 149,534- molecule unique training set into 49,660 unique generated strings, retained 40,477 chemically filtered molecules, completed a five-conformer structural screen for 500 candidates, and executed 14,259 real Vina counter-screening jobs. The analysis then reduced the data to a uniform 147×97 matrix without filling missing jobs with synthetic scores. These are practical successes in reproducible pipeline construction, data provenance, and large-scale RNA-screen orchestration.

The strongest candidate-level result is target-score enrichment within the generated population. Every one of the 40 ranked candidates received a more favorable mean *SMN2* score than the median of ten property-matched controls, with a mean gap of 0.729 kcal/mol. The result exceeded all 2,000 random cohorts, and the low pairwise fingerprint similarity and 977 scaffold labels among 1,000 exported molecules show that the workflow did not reach this stage by reproducing a narrow set of near-duplicate structures. Because a target-score-only cohort enriched at least as strongly, the result validates the pipeline's ability to enrich its own population for favorable target scores; it does not establish an incremental benefit for the complete ranking rule or prove binding.

The hard-decoy stage adds value by preventing that target-score result from being overinterpreted. Across the top ten candidates, the weighted mean gap spans only 0.44 kcal/mol, whereas the conservative gap spans 2.24 kcal/mol and is negative throughout. The same worst-case pattern occurs across all 147 evaluable molecules and under the two held-out splicing-modifier controls. Rather than treating this as a failed pipeline, it is more accurate to treat it as a successful quality gate: the system detected that average target–decoy separation did

TABLE III
REDOCKING RECOVERY AGAINST EXPERIMENTALLY DETERMINED BOUND POSES

PDB	Ligand	Rank-1 RMSD vs. model 1 (Å)	Rank-1 RMSD ensemble min. (Å)	Best sampled RMSD (Å)	Rank of best mode	Contact recall	Contact Jaccard
8R63	Branaplam	1.307	1.244	1.137	2	0.810	0.630
8R8P	SMN-CX	7.169	6.986	1.856	20	0.273	0.162
5KX9	Ribocil analogue	6.173	6.173	4.384	15	0.271	0.198
8R62	Risdiplam	12.587	11.923	2.478	14	0.000	0.000

not survive the strongest member of a bias-matched panel. Prior work has shown that RNA-targeting models require careful negative examples to avoid shortcut learning [12]; the present architecture operationalizes that principle for structural screening.

Redocking provides the necessary calibration for interpreting this gate. Branaplam was recovered at rank 1 with 1.307 Å model-1 RMSD and 81% contact recall, demonstrating that the fixed workflow can recover an experimental RNA–ligand pose in at least one relevant case. For SMN-CX and risdiplam, lower-RMSD modes were sampled but not ranked first. Thus, the benchmark separates a sampling capability from a scoring limitation and identifies a concrete improvement path: preserve the sampled ensemble and add RNA-aware reranking. Until that is done, negative conservative gaps characterize this Vina protocol, not the biological selectivity of the molecules.

The methodological contribution is therefore the screening architecture and its audit trail. Selectivity is evaluated in the same workflow as target scoring rather than as a post-hoc claim; worst-case and average separation are reported together; the target is represented by five conformers rather than one structure; known modifiers are held out from training; and missing jobs remain explicit. Desirability functions use fixed prespecified windows rather than comparison-set min–max normalization, so adding or removing a compound does not change the meaning of an acceptance level. These choices are consistent with RNA-targeted drug-design strategies that combine structural, chemical, and interaction evidence rather than relying on one score alone [6], [8].

The SRSI ablation further clarifies what the integrated ranking does. The full ranking correlated 0.822 with target score alone and shared 33 of its top 40 molecules with the target-only ranking, while individual rank shifts reached 103 positions. The composite therefore is neither identical to target-only selection nor independently validated. It is a transparent prioritization policy whose added value must ultimately be tested against experimental labels. Likewise, attribution of a linear standardized score primarily documents configured weights and feature dispersion; it should not be presented as biological mechanism.

A. Limitations

This study is entirely computational. Vina scores, predicted rescue scores, desirability values and SRSI values are prioritization metrics, not measurements of exon 7 inclusion, RNA-binding affinity, cellular activity, toxicity or therapeutic efficacy. The off-target risk index is an index defined by this workflow, not a probability of biological off-target activity.

The docking protocol was tested, not biologically validated. Section III-H reports rank-1 recovery for one of four benchmark complexes. This bounds every quantity in Sections III-C, III-D and III-E: the Vina scores, weighted and conservative gaps, off-target risk index, desirability aggregates, and matched-control enrichment describe this scoring function and cannot be read as RNA binding or selectivity measurements. The direction of bias is unknown. AutoDock Vina was parameterized on protein–ligand complexes, and the cited assessments report limited RNA–ligand accuracy [14], [15]. The five target conformers are the first five deposited models of the 8R62 NMR ensemble; conformer- level persistence of candidate contacts was not evaluated.

The splicing-rescue and contact quantities are explicit engineering proxies, but they are not biologically calibrated. In particular, beneficial and caution counts are fixed fractions of the global close- contact count when corresponding landmark labels are present, rather than nucleotide-specific contact assignments. The reported minimum U1 distance is also the global ligand–receptor minimum when a U1 label is present. A minimum of 0.970 Å may therefore reflect a steric clash, not productive recognition. Pose support approaches its ceiling and the occlusion term varies by only 7.5×10^{-4}, so these components have limited discriminatory value despite their configured weights. The ablation records rank effects but does not validate mechanism.

Only the ten highest-ranked candidates are tabulated in Section III-C, but the worst-case claim is not an extrapolation from them: the conservative gap was evaluated for every evaluable molecule and is reported in Section III-C. The ranking rule itself could not have supported such an inference, because the lexicographic order is settled in practice by the weighted gap alone, so rank position carries no information about the conservative gap.

Raw Vina scores were compared across receptors of differing size and composition without ligand-efficiency normalization. Because the score scales with ligand heavy-atom count, comparisons between molecules of different size, and selectivity gaps computed by subtracting scores obtained on different receptors, are both susceptible to size artifacts. The apparent target-score advantage of MolGen_038292 over two substantially smaller reference compounds should be regarded as unresolved until efficiency-normalized scores are reported.

The sequence-level decoy atlas was specified at over 180,000 motifs, but only 100 decoys were built as three-dimensional receptors, and 97 were available in the uniform analyzed matrix. Equal coverage across 147 molecules establishes comparability within this pilot panel; it does not

establish transcriptome-wide selectivity. The strongest-decoy statistic is also panel-size dependent because adding receptors creates more opportunities for an extreme score. Sensitivity across independent panels and multiple panel sizes remains necessary.

The 400 candidate–control assignments used only 183 distinct controls, so control reuse creates dependence not represented by a simple candidate-level sign-flip test. The random-cohort and target-only analyses address selection effects but do not supply an external chemical-library control. Cluster-aware uncertainty analysis and an external benchmark are still needed.

The generative model is unconditional and was trained on ChEMBL/ZINC subsets; the configured R-BIND and SMM-RNA sources contributed zero rows. The run record did not capture the exact database release identifiers, download dates, or source licenses, and novelty was approximated against 20,000 training fingerprints rather than the complete corpus. No controlled n-gram, GRU, or library-sampling baseline is reported. These limitations prevent attribution of screening performance to the Transformer architecture or to RNA-ligand-focused training.

The most direct next step is RNA-aware reranking of the retained pose ensembles: Table III shows that lower-RMSD modes for risdiplam and SMN-CX were already sampled. A strengthened study should also maintain a prespecified benchmark of at least 20–30 diverse RNA–ligand complexes, report ligand-efficiency and receptor-normalized scores, repeat the hard-decoy analysis across panel sizes, and add experimental *SMN2* splicing, direct-binding, and cellular assays [1], [2].

V. Conclusion

SMN2-RescueDiff implements a reproducible multi-stage computational workflow combining unconditional autoregressive molecular generation, chemical-quality and diversity filtering, five-conformer *SMN2* docking, staged human-RNA decoy counter-screening, held-out controls, and an ablatable multi-factor ranking scheme. The audited run produced 40,477 filtered generated molecules, exported 1,000 diverse docking candidates, completed all 2,500 target jobs, and assembled 14,259 real-Vina decoy results into a common 147×97 comparison matrix. This establishes that the complete computational workflow is executable, traceable, and capable of narrowing a large generated library without silently imputing missing evidence.

The pipeline also produced a reproducible positive screening signal: all 40 leading candidates had more favorable mean target scores than their property-matched control medians, with 0.729 kcal/mol mean enrichment and no equally large result among 2,000 random cohorts. That enrichment is best interpreted as evidence that the staged workflow can select favorable target scores within its generated population. It is not proof of binding, and a target-only ranking performed at least as well, so the added biological value of SRSI remains to be validated.

The hard-decoy and redocking modules strengthen the study by defining where interpretation must stop. Favorable average target–decoy gaps did not translate into worst-case separation, and only branaplam achieved sub-2 Å rank-1 recovery in the four-complex redocking set. At the same time, lower-RMSD modes were sampled for SMN-CX and risdiplam, providing a concrete route toward RNA-aware reranking rather than a need to discard the pipeline. The resulting contribution is an auditable candidate-prioritization architecture that produces testable molecular hypotheses while making its calibration limits visible.

The identified molecules remain *in silico* candidates. No claim is made that they bind *SMN2*, increase exon 7 inclusion, improve cellular phenotypes, are safe, or outperform risdiplam or branaplam. Those questions require orthogonal computational scoring and prospective experimental validation. Within that boundary, SMN2-RescueDiff provides a reproducible foundation for the next round of RNA-aware rescoring and laboratory prioritization.

Acknowledgments

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Code and Data Availability

The exact audited snapshot is commit `deacf03` on the `real-pipeline-rigor` branch of the project repository, github.com/sakshams-cyber/SMN2-RescueDiff. The `reproducibility/tcbb_2026` release contains the 100 decoy receptors, five target receptors, prepared ligand PDBQT files, complete score tables, per-job status records, the 741-job missingness table, generation audit, Stage C.5 inputs, Module 5 ablation outputs, four- complex redocking structures and tables, source manifest, and SHA-256 checksums. The release validation records Python 3.11.15, AutoDock Vina 1.2.7, Open Babel 3.2.1, and spyrmsd 0.9.0. Figure-generation code and the manuscript source are included with this revision package.

Declarations

Ethical approval: Not applicable. This study is entirely computational and involved no human participants, human data or animal subjects.

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Saksham Srivastava is a student at Saint Francis High School, Mountain View, CA, USA. He works on computational biology and machine learning for RNA-targeted drug design, and developed the SMN2-RescueDiff pipeline described here.